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Magnetic susceptibility and crystal structure analysis of potassium iron oxalate trihydrate

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Abstract

The compound with formula $K_3Fe(C_2O_4)_3 \cdot 3H_2O$ has been studied by magnetic susceptibility and X-ray powder diffraction. The magnetic susceptibility data measured in the temperature range 80–300 K obey the Curie–Weiss law with a negative paramagnetic Curie temperature: $\theta = -3.0(1)$ K, suggesting a predominantly weak antiferromagnetic interaction between iron ions in the structure. The effective magnetic moment shows a value of $5.88\mu_B$, which is slightly lower than the spin-only value of $5.9\mu_B$ for spin of $S = \frac{5}{2}$ (Fe^{+3}). X-ray powder diffractometry was used to investigate the structure of this compound. It has been found that it crystallizes in the monoclinic space group $P2_1/c$, with unit cell parameters $a = 7.762(1)$ Å, $b = 19.937(3)$ Å, $c = 10.353(1)$ Å, and $\beta = 107.75(1)^\circ$. © 2002 Elsevier Science B.V. All rights reserved.

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1. Introduction

The salts based on oxalate complexes are interesting materials, which have technological applications as precursors to nanocrystalline metallic oxides [1,2], and to molecular-based magnetic materials [3–6]. In these novel compounds, the oxalate ion can act as bridging ligand between metals and thus constructing two- or three-dimensional honeycomb anionic networks $[M_2(C_2O_4)_3]_\infty$ that behave as ferro [5,6], ferri [3,6] or antiferro-magnets [4,6]. The crystal structures of trioxalate metallate salts of general formula M^I_3

$M^{III}(C_2O_4)_3 \cdot 3H_2O$, with $M^I = K, Rb, NH_4$ and $M^{III} = Al, Cr, Ga, Fe, Mn$ [7,8] have been reported as monoclinic cells, space group $P2_1/c$, or as triclinic cells, space group $P\bar{1}$. The structures consist of $M^{III}(C_2O_4)_3^-$ moieties, with M^{III} atoms coordinated octahedrally to three oxalate groups acting as bidentate ligands [9]. Among these compounds, the one with formula $K_3Fe(C_2O_4)_3 \cdot 3H_2O$ is found as a natural mineral named *minguzitte*, and can be prepared in the laboratory by a two-step synthesis. Aiming to prepare metallic oxalate with novel molecular structures and potential magnetic properties, we have undertaken the characterization by magnetic susceptibility and X-ray diffraction of the precursor compound $K_3Fe(C_2O_4)_3 \cdot 3H_2O$.

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2. Experimental

The complex $\text{K}_3\text{Fe}(\text{C}_2\text{O}_4)_3 \cdot 3\text{H}_2\text{O}$ was prepared as described elsewhere [9]. High temperature magnetic susceptibility data was obtained by means of an automated Faraday system in the temperature range 80–300 K. Powder diffraction data was obtained with a Bruker AXS diffractometer, using Ni-filtered $\text{CuK}\alpha$ radiation at room temperature in θ/θ reflection mode. The pattern was scanned over the angular range $7\text{--}80^\circ$ (2θ) with a step size of 0.02° and counting time of 40 s. Quartz was used as an external standard.

3. Results and discussion

The DC magnetic susceptibility measurements in the temperature range 80–300 K is shown in Fig. 1. It is possible to note that temperature dependence of inverse magnetic susceptibility gives a straight line in the whole temperature range. From a good fit of the data to the Curie–Weiss law $\chi = C/T - \theta$ was found for the Curie constant $C = 9.908 \times 10^{-3}$ (emu K/g) and for the paramagnetic Curie temperature θ was obtained: $\theta = -3.0(1)$ K. From the C value it was possible to calculate the effective magnetic moment for the iron ions by using $C = N\mu_{\text{eff}}^2/3k_B$, the obtained value $\mu_{\text{eff}} = 5.88\mu_B$, which is slightly lower than

the spin-only value of $5.9\mu_B$ for spin of $S = \frac{5}{2}$ (Fe^{+3}).

The negative sign of the θ Curie temperature suggest the presence of predominant anti-ferromagnetic interactions between the Fe localized magnetic moments. The small negative θ value could be due to zero field splitting of the iron(III) ions, or to dipolar magnetic interactions between anions. This last possibility should not be significant since the iron(III) ions are more than 7 Å apart from each other. The Fe–Fe distance obtained from the structural analysis was 7.021(8) Å.

The X-ray powder diffraction pattern was indexed in a monoclinic system using the auto-indexing program DICVOL91 [10], with unit cell parameters: $a = 7.762(1)$ Å, $b = 19.937(3)$ Å, $c = 10.353(1)$ Å, $\beta = 107.75(1)^\circ$, and figures of merit $M_{20} = 30.3$ [11] and $F_{30} = 62.1$ (0.0078, 62) [12]. The systematic absences indicated that the compound crystallizes in space group $\text{P2}_1/\text{c}$.

A Rietveld refinement [13], using as starting model for $\text{K}_3\text{Fe}(\text{C}_2\text{O}_4)_3 \cdot 3\text{H}_2\text{O}$ the isostructural compound $\text{K}_3\text{Cr}(\text{C}_2\text{O}_4)_3 \cdot 3\text{H}_2\text{O}$ [14], converged to the final figures of merit: $R_P = 11.61\%$, $R_{WP} = 4.50\%$, $R_{\text{exp}} = 3.06\%$ and $\chi^2 = 1.47$. The peak profiles were modeled using a pseudo-Voigt peak-shaped function. The diffraction pattern and the difference between the measured and calculated pattern are shown in Fig. 2.

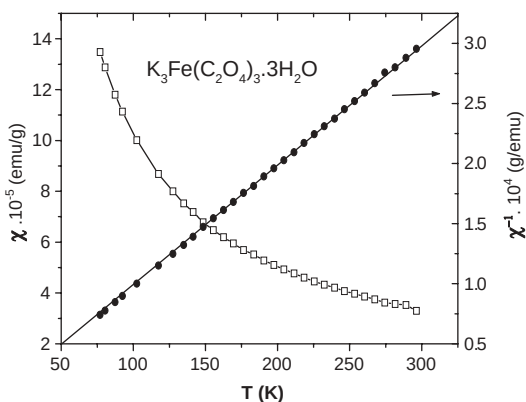


Fig. 1. Magnetic susceptibility and its inverse as a function of temperature for $\text{K}_3\text{Fe}(\text{C}_2\text{O}_4)_3 \cdot 3\text{H}_2\text{O}$.

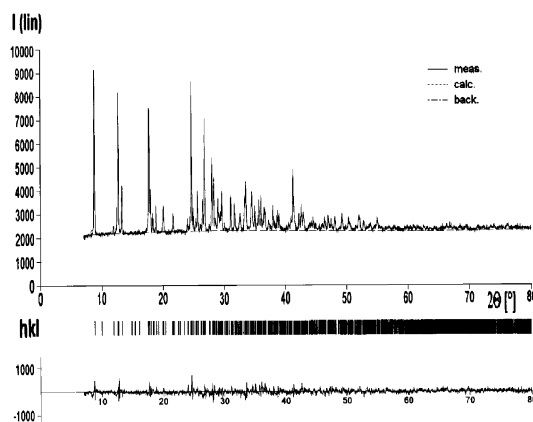


Fig. 2. Final Rietveld refinement plot showing the difference between the observed (·) and calculated (-) diffraction patterns of the complex $\text{K}_3\text{Fe}(\text{C}_2\text{O}_4)_3 \cdot 3\text{H}_2\text{O}$.

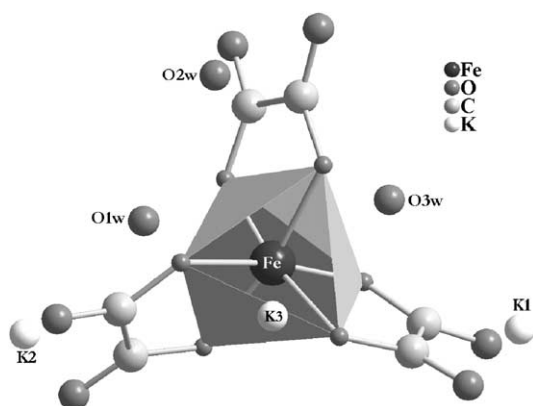


Fig. 3. View of the structure of $\text{K}_3\text{Fe}(\text{C}_2\text{O}_4)_3 \cdot 3\text{H}_2\text{O}$ showing the octahedral coordination around the iron atom.

The crystal structure consists of discrete $\text{Fe}(\text{C}_2\text{O}_4)_3^{3-}$ anions, K^+ cations and water molecules. Six oxygen atoms from water molecules and oxalate anions surround each of the K^+ atoms at distances varying from 2.60(6) to 2.90(9) Å. The structure of the tri(bidentate) $\text{Fe}(\text{C}_2\text{O}_4)_3^{3-}$ complex shows a distorted octahedral coordination with angles O–Fe–O between 81.03(1)° and 171.89(2)°, and Fe–O distances varying between 1.978(7) and 2.002(7) Å. Fig. 3 shows a view of the metal complex with the octahedral coordination around the iron atom.

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